Name: ______________________ **Date:** ______________

Introduction to Pythagoras' Theorem

Year 8 — Stage 4 Mathematics

Learning Intentions

- I can identify the hypotenuse in a right-angled triangle.
- I can use Pythagoras' theorem to calculate the hypotenuse.
- I can rearrange Pythagoras' theorem to find a shorter side.
- I can apply Pythagoras' theorem to a real-world situation.

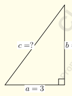
Key Rule

In a right-angled triangle, the **hypotenuse** c (longest side, opposite the right angle) satisfies:

$$c^2 = a^2 + b^2 \quad \Longrightarrow \quad c = \sqrt{a^2 + b^2}$$

To find a **shorter side**: $a^2 = c^2 - b^2 \quad \Longrightarrow \quad a = \sqrt{c^2 - b^2}$

Worked Example — Finding the Hypotenuse



Step 1: Write the theorem.

$$c^2 = a^2 + b^2$$

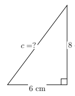
Step 2: Substitute and evaluate.

$$c^2 = 3^2 + 4^2 = 9 + 16 = 25$$

Step 3: Square root both sides.

$$c = \sqrt{25} = \mathbf{5}$$

1. [**Fluency**] Find the length of the hypotenuse c in the triangle below. Show all working.



2. [**Misconception Check**] A right-angled triangle has a hypotenuse of 13 cm and one shorter side of 5 cm.

   (a) Find the length of the unknown shorter side a. Show full working.

   (b) A student attempted the same problem and wrote:

   $$a^2 = 5^2 + 13^2 = 25 + 169 = 194, \qquad \text{so} \quad a = \sqrt{194} \approx 13.9 \text{ cm}.$$

   Identify the exact error the student made. Then write the correct solution.

3. [**Challenge — Real-World Application**] A 5 m ladder leans against a vertical wall. The foot of the ladder sits on level ground, 1.8 m from the base of the wall.

   (a) Draw and label a right-angled triangle to represent this situation. Mark the hypotenuse, both known measurements, and the unknown height h.

   (b) Calculate h, the height the ladder reaches up the wall. Give your answer correct to one decimal place.

   (c) The foot of the ladder is then moved to 2.5 m from the wall. The ladder length stays the same.
   Without fully recalculating, **explain** whether the ladder now reaches *higher* or *lower* up the wall, and why.

Teacher Answer Key

For teacher use only — not for student distribution

1. Fluency — Finding the hypotenuse (6–8–10 triple)

$$\begin{aligned} c^2 &= 6^2 + 8^2 \\ &= 36 + 64 \\ &= 100 \\ c &= \sqrt{100} = \mathbf{10} \text{ cm} \end{aligned}$$

Marking: 1 mark — correct substitution into $c^2 = a^2 + b^2$. 1 mark — correct answer $c = 10$ cm.

2. Misconception Check — Finding a shorter side (5–12–13 triple)

(a)

$$\begin{aligned} a^2 &= c^2 - b^2 \\ &= 13^2 - 5^2 \\ &= 169 - 25 \\ &= 144 \\ a &= \sqrt{144} = \mathbf{12} \text{ cm} \end{aligned}$$

Marking: 1 mark — correct rearrangement $a^2 = c^2 - b^2$. 1 mark — correct answer $a = 12$ cm.

(b) The student **added** the squares $(5^2 + 13^2)$ instead of **subtracting** $(13^2 - 5^2)$. When finding a shorter side, the theorem must be rearranged to $a^2 = c^2 - b^2$. A further red flag: the student's answer ($\approx$13.9 cm) is *larger* than the hypotenuse (13 cm), which is geometrically impossible. Correct answer: $a = 12$ cm.

Marking: 1 mark — correctly identifies addition error. 1 mark — corrected working and answer.

IMPORTED FROM CHAT:
[TEACHER ONLY] Refinement suggestion — add student-facing mark allocations. Currently no marks appear beside each part. For NSW Stage 4 exam readiness, consider annotating each part with (__/2) or similar so students practise gauging how much to write. Q2(b) in particular could become a 2-mark response question with an explicit instruction to give two distinct points (name the error; explain why the answer is impossible). This also doubles as a useful self-assessment prompt.

Claude Chat: *https://claude.ai/share/84170245-13fd-472f-a116-876fc87809b7*

Name: ______________________ **Date:** ______________

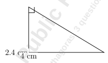
Introduction to Pythagoras' Theorem

Key Idea: In a right-angled triangle, the square of the hypotenuse equals the sum of the squares of the other two sides.

$$c^2 = a^2 + b^2$$

1. A right-angled triangle has legs $a = 3$ cm and $b = 4$ cm. Find the length of the hypotenuse c.

2. The triangle below is right-angled at B. Use the diagram to find the length of the hypotenuse.



3. A student claims that if $a = 5$ and $b = 12$, then the hypotenuse is $c = 17$ because "you just add the sides". Explain why this is incorrect and calculate the correct value of c.

Teacher Answer Key

4. Solution:

$$c^2 = 3^2 + 4^2 = 9 + 16 = 25$$

$$c = 5 \text{ cm}$$

Marking guidance: correct substitution; correct square/sum; correct square root.

5. Solution:

$$c^2 = 2.4^2 + 4^2 = 5.76 + 16 = 21.76$$

$$c = \sqrt{21.76} \approx 4.66 \text{ cm}$$

Marking guidance: diagram interpreted correctly; right angle identified; rounding acceptable to two decimal places.

6. Solution: The claim $c = a + b$ is incorrect because Pythagoras' Theorem uses squares, not addition.

$$c^2 = 5^2 + 12^2 = 25 + 144 = 169$$

$$c = 13$$

Marking guidance: clear explanation of misconception; correct working; correct final value.

IMPORTED FROM CHAT:
— [TEACHER ONLY] — Consider adding one more representation-shift item (e.g., a rotated triangle) to strengthen transfer.

Copilot Chat:
https://copilot.microsoft.com/shares/FMudwG7axsTVEF3UnXSeY

Name: ______________________ **Date:** ______________

Year 8: Introduction to Pythagoras' Theorem

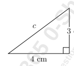
Pythagoras' Theorem applies to right-angled triangles.

$$a^2 + b^2 = c^2$$

where c is the hypotenuse (the longest side).

1. Getting Started

The triangle below is right-angled.



Use Pythagoras' Theorem to find the value of c.

2. Building Confidence

A right-angled triangle has side lengths 5 cm and 12 cm.

Find the hypotenuse.

3. Challenge

The hypotenuse of a right-angled triangle is 13 cm.

One shorter side is 5 cm.

Find the length of the remaining side.

Teacher Answer Key

4.

$$c^2 = 4^2 + 3^2$$

$$c^2 = 16 + 9 = 25$$

$$c = 5 \text{ cm}$$

Marking guidance: Substitutes correctly into the formula and calculates $c = 5$ cm.

5.

$$c^2 = 5^2 + 12^2$$

$$c^2 = 25 + 144 = 169$$

$$c = 13 \text{ cm}$$

Marking guidance: Identifies the hypotenuse and applies the theorem correctly.

6.

$$13^2 = 5^2 + b^2$$

$$169 = 25 + b^2$$

$$b^2 = 144$$

$$b = 12 \text{ cm}$$

Marking guidance: Rearranges correctly and solves for the unknown side.

IMPORTED FROM CHAT:
— [TEACHER ONLY] — Add a fourth "spot the error" question where a student incorrectly calculates $3^2 + 4^2 = 7^2$ to explicitly target a common Year 8 misconception about squaring numbers.

Name: ______________________ **Date:** ______________

Year 8 Introduction to Pythagoras' Theorem

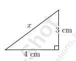
Key Rule
For a right-angled triangle with shorter sides a and b, and longest side c:

$$a^2 + b^2 = c^2$$

The longest side opposite the right angle is called the hypotenuse.

1. Starter

Find the length of the hypotenuse.



Find x.

2. Practice

A right-angled triangle has a hypotenuse of 13 cm and one shorter side of 5 cm.

Find the length of the other side.

3. Challenge

A ladder is leaning against a wall. The foot of the ladder is 6 m from the wall and the ladder itself is 10 m long.

How high up the wall does the ladder reach?

Give your answer in metres.

Teacher Answer Key

1.

$$x^2 = 4^2 + 3^2$$

$$x^2 = 16 + 9$$

$$x^2 = 25$$

$$x = 5$$

Answer: 5 cm

Marking guidance:

- Substitutes correctly into Pythagoras' theorem.
- Squares numbers correctly.
- Gives answer with units.

2.

$$13^2 = 5^2 + b^2$$

$$169 = 25 + b^2$$

$$b^2 = 144$$

$$b = 12$$

Answer: 12 cm

Marking guidance:

- Identifies the hypotenuse.
- Rearranges correctly.
- Gives answer with units.

3.

$$10^2 = 6^2 + h^2$$

$$100 = 36 + h^2$$

$$h^2 = 64$$

$$h = 8$$

Answer: 8 m

Marking guidance:

- Forms the correct equation.
- Solves accurately.
- Includes units.

IMPORTED FROM CHAT:
To draft the worksheet, I need one detail: what is the topic (and year/stage) you want the worksheet to cover? For example: Year 9 Maths – Linear Relationships, Stage 5 Science – Chemical Reactions, or Year 11 Maths Advanced – Trigonometric Functions.

— [TEACHER ONLY] — Consider adding a misconception-check question where students must identify the hypotenuse before applying Pythagoras' theorem, as Year 8 students commonly square the wrong side.

ChatGPT Free: *https://chatgpt.com/share/6a3b1963-6838-83ec-85e5-eebf5448db33*

Name: ______________________ **Date:** ______________

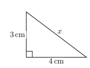
Introduction to Pythagoras' Theorem

Learning Intention

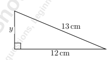
To understand and apply Pythagoras' Theorem ($c^2 = a^2 + b^2$) to find missing side lengths in right-angled triangles. Remember that c must always be the **hypotenuse** (the longest side, directly opposite the right angle).

1. Fluency: Finding missing sides
For each of the right-angled triangles below, use Pythagoras' Theorem to calculate the value of the unknown side. Show all steps of your working.

a) Find the hypotenuse, x.



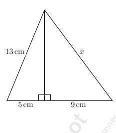
b) Find the shorter side, y.



2. Application: The Farm Gate
A farmer is building a rectangular wooden gate that is 2.4 m wide and 1.0 m high. To ensure the gate does not sag, she needs to attach a diagonal brace across the back.

(a) Draw a simple, labelled diagram to represent this scenario.
(b) Calculate the exact length of the diagonal piece of wood required.

3. Challenge: The Composite Shape
The diagram below shows two right-angled triangles joined together. Calculate the length of the side marked x.
Hint: You will need to find the length of the shared middle side first.



IMPORTED FROM CHAT:
I am ready to design your LaTeX worksheet. Could you please provide your teacher brief, including the subject (Maths, Science, or Integrated STEM), Year/Stage, topic, and preferred pedagogy mode?

— [TEACHER ONLY] — Refinement Suggestion: If you notice students struggling with identifying whether to add or subtract squares, I can add a quick "Misconception Check" diagnostic question before the fluency task where they must spot and explain the error in a provided, flawed calculation.

Gemini chat:
https://gemini.google.com/share/d/1OEdOeqMA57jSFht7kKp3kr2ZtA5Pa0Lq?usp=sharing

Teacher Answer Key

Question 1: Fluency

- **1a)** Using $c^2 = a^2 + b^2$
  $x^2 = 3^2 + 4^2$
  $x^2 = 9 + 16$
  $x^2 = 25$
  $x = \sqrt{25} = 5\,\text{cm}$
  Marking guide: 1 mark for correct substitution, 1 mark for correct final answer with units.
- **1b)** Using $a^2 + b^2 = c^2$
  $y^2 + 12^2 = 13^2$
  $y^2 + 144 = 169$
  $y^2 = 169 - 144$
  $y^2 = 25$
  $y = \sqrt{25} = 5\,\text{cm}$
  Marking guide: 1 mark for identifying the hypotenuse is 13 and setting up the subtraction/equation correctly, 1 mark for final answer.

Question 2: Application

- **2a)** Students should draw a rectangle with base 2.4 m, height 1.0 m, and a diagonal drawn in. A right-angle symbol should be present in the corner.
- **2b)** Using $c^2 = a^2 + b^2$
  $c^2 = 2.4^2 + 1.0^2$
  $c^2 = 5.76 + 1.0$
  $c^2 = 6.76$
  $c = \sqrt{6.76} = 2.6\,\text{m}$
  Marking guide: 1 mark for diagram, 1 mark for calculation, 1 mark for final answer in metres.

Question 3: Challenge

- **Step 1: Find the shared vertical side (let's call it h)**
  Looking at the left triangle:
  $h^2 + 5^2 = 13^2$
  $h^2 + 25 = 169$
  $h^2 = 144$
  $h = \sqrt{144} = 12\,\text{cm}$
- **Step 2: Find x**
  Looking at the right triangle:
  $x^2 = h^2 + 9^2$
  $x^2 = 12^2 + 9^2$
  $x^2 = 144 + 81$
  $x^2 = 225$
  $x = \sqrt{225} = 15\,\text{cm}$
  Marking guide: 1 mark for finding the intermediate side (12cm), 1 mark for setting up the second equation, 1 mark for correct final answer.

Name: ______________________ **Date:** ______________

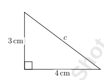
Introduction to Pythagoras' Theorem

Key idea: In a right-angled triangle, the square of the hypotenuse equals the sum of the squares of the other two sides:

$$a^2 + b^2 = c^2$$

1. Find the hypotenuse

A right-angled triangle has side lengths 3 cm and 4 cm.



Find the length of the hypotenuse.

2. Find a missing side

A right-angled triangle has hypotenuse 13 cm and one side 5 cm.

Find the length of the other side.

3. Challenge: Word problem

A ladder leans against a wall. The base of the ladder is 6 m from the wall, and the ladder is 10 m long.

- How high up the wall does the ladder reach?

Teacher Answer Key

1.

$$c^2 = 3^2 + 4^2 = 9 + 16 = 25$$
$$c = \sqrt{25} = 5\,\text{cm}$$

Answer: 5 cm

2.

$$c^2 = a^2 + b^2$$
$$13^2 = 5^2 + b^2$$
$$169 = 25 + b^2$$
$$b^2 = 144$$
$$b = \sqrt{144} = 12\,\text{cm}$$

Answer: 12 cm

3.

$$c^2 = a^2 + b^2$$
$$10^2 = 6^2 + h^2$$
$$100 = 36 + h^2$$
$$h^2 = 64$$
$$h = \sqrt{64} = 8\,\text{m}$$

Answer: 8 m

IMPORTED FROM CHAT:
I'm ready to build this, but I need a bit more detail first. What are the key inputs: subject (Maths/Science), year or stage, topic, number of questions, and desired difficulty or mode (e.g. exam-style, scaffolded, investigation)?

— [TEACHER ONLY] —
Add a misconception check (e.g. a triangle that is not right-angled) to ensure students do not apply Pythagoras indiscriminately.

Perplexity Chat:
https://www.perplexity.ai/search/90e8d150-9962-463f-8268-ee5ea7ac3962